# Yuzong Chen

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## EDUCATION

- **Cornell University** Ithaca, NY, USA
  Ph.D. in Electrical and Computer Engineering *Aug. 2022 - May. 2027 (Expected)*
  *Advisor: Prof. Mohamed Abdelfattah*
- **Nanyang Technological University (NTU)** Singapore
  B.Eng. in Electrical & Electronic Engineering *Aug. 2015 - Jun. 2019*
  GPA: 4.74 / 5.00, Honours (Highest Distinction)

## RESEARCH EXPERIENCE

- **Computer Systems Laboratory (CSL),** Cornell University USA
  *Graduate Research Assistant (Advisor: Prof. Mohamed Abdelfattah)* *Aug. 2022 - Present*
  **Efficient FPGA Architectures for Machine Learning:** I am currently investigating novel FPGA architectures for accelerating deep neural networks.
- **Centre for Integrated Circuits and Systems (CICS),** Nanyang Technological University Singapore
  *Project Officer (Advisor: Prof. Tony Tae-Hyoung Kim)* *Feb. 2020 - Aug. 2021*
  **Computing In-Memory:** I conducted projects about computing in-memory (CIM) circuit design based on static random access memory (SRAM) and resistive random access memory (ReRAM). My projects mainly explored how to perform versatile CIM functions on a single memory chip to save area and improve flexibility.
- **Department of Electrical and Computer Engineering,** National University of Singapore Singapore
  *Research Engineer (Advisor: Prof. Heng Chun Huat)* *Sept. 2021 – Jul. 2022*
  **RF Digital Power Amplifier:** I worked on a joint project with NXP Semiconductors to design an RF switched-capacitor power amplifier for high-speed communication. I implemented several key circuit blocks and helped with the chip tapeout in 22nm FDSOI technology.

## PUBLICATIONS

- **Journal Articles**

[1] Donghyuk Kim, Chengshuo Yu, Shanshan Xie, Yuzong Chen, Joo-Young Kim, Bongjin Kim, Jaydeep Kulkarni, Tony Tae-Hyoung Kim, "An Overview of Processing-in-Memory Circuits for Artificial Intelligence and Machine Learning," in *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)*, 2022.

[2] Yuzong Chen, Lu Lu, Bongjin Kim, Tony Tae-Hyoung Kim, "A Reconfigurable 4T2R ReRAM Computing In-Memory Macro for Efficient Edge Applications," in *IEEE Open Journal of Circuits and Systems (OJCAS)*, vol. 2, pp. 210-222, 2021.

[3] Yuzong Chen, Lu Lu, Bongjin Kim, Tony Tae-Hyoung Kim, "Reconfigurable 2T2R ReRAM Architecture for Versatile Data Storage and Computing In-Memory," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 28, no. 12, pp. 2636-2649, Dec. 2020.

- **Conference Proceedings**

[1] Yuzong Chen, Junjie Mu, Hyunjoon Kim, Lu Lu, Tony Tae-Hyoung Kim, "A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2022.

[2] Yuncheng Lu, Zehao Li, Yuzong Chen, Tony Tae-Hyoung Kim, "A 181μW Real-Time 3-D Hand Gesture Recognition System based on Bi-directional Convolution and Memoryless Clustering," in *IEEE Custom Integrated Circuits Conference (CICC)*, 2022.

[3] Yuzong Chen, Lu Lu, Yuncheng Lu, Tony Tae-Hyoung Kim, "A Multi-Functional 4T2R ReRAM Macro Enabling 2-Dimensional Access and Computing In-Memory," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021.

[4] Lu Lu, Yuzong Chen, Tony Tae-Hyoung Kim, "A Configurable Randomness Enhanced RRAM PUF with Biased Current Sensing Scheme," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021.

[5] Vishal Sharma, Ju Eon Kim, Yong-Jun Jo, Yuzong Chen, Tony Tae-Hyoung Kim, "AND8T SRAM Macro with Improved Linearity for Multi-bit In-Memory Computing," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021.

[6] Yuzong Chen, Lu Lu, Bongjin Kim, Tony Tae-Hyoung Kim, "Reconfigurable 2T2R ReRAM with Split Word-lines for TCAM Operation and In-Memory Computing," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2020.

- **Book Chapters**

[1] Tony Tae-Hyoung Kim, Yuzong Chen, Lu Lu, "ReRAM-based Processing-in-Memory (PIM)," in *Processing-in-Memory for AI from Circuits to Systems*, Springer, 2022, pp. 93-120.

## EDITORIAL SERVICE

- **Journal Reviewer:** IEEE Transactions on Very Large Scale Integration Systems (TVLSI); IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I); IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS). IEEE Open Journal of Circuits and Systems (OJCAS).

## HONOURS AND AWARDS

- NTU Dean's List (top 5% of the cohort) in academic year 15-16, 16-17
- NTU Science and Engineering Undergraduate Scholarship
- Excellent Internship Award for the professional internship with ExxonMobil

## TECHNICAL SKILL

- **Programming Languages**
  Python, C++, MATLAB, Verilog.
- **Computer Software Tools**
  Cadence Virtuoso, Xilinx Vivado, Git, TensorFlow.